

How Retrieval Quality Depends on Document Ingestion

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1 Introduction

Retrieval quality is decided twice: once while the document is read and once while the index is searched. A chunk cut in the wrong place cannot be recovered by a better ranker, so the ingestion stage is part of the result.

A chunk is the unit of retrieval that the generator reads.

Figure 1: Mean recall at five for each architecture on domain one.

Architecture	Dense	Sparse	Hybrid
Dense	0.61	0.00	0.68
Sparse	0.34	0.72	0.70
Rerank	0.55	0.69	0.74

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2 Ingestion and Cleaning

Text taken from a PDF arrives in pieces. A word split across two lines arrives as a hyphen at the end of one line and a fragment at the start of the next, and the word has to be put back together before the reader can judge it: the score is mea-
sured by the annotators and not by the parser.

A chunk is the unit of retrieval that the generator reads.

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The stamp above is not a heading. A layout model sometimes reads the margin as one, and the text under it still belongs to the section above it.

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3 Results and Findings

Hybrid retrieval came first on both domains, and the gap over dense retrieval was wider on the domain with the more unusual vocabulary. Reranking on top of hybrid retrieval added little, which is reported here rather than left out.

1 Correspondence to retrieval.study@example.org

2 The loader is unchanged between the two runs reported here.

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Reading order is what a layout model guesses at and what the rest of the project then depends on, so the guess is worth pinning down.

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Nothing on this page starts a new section, which is the point of it: the last section of the document carries on to the end, and a page boundary in the middle of it is not a section boundary.

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